

GROUP - 7

“CorText: A Semantic-Driven AI for Personalized Medicine Recommendation in Clinical Management System”

Group 7 – Member Assignment

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Section	Assigned Members	Focus
Background of the Study	1. CHRISTOPHER O. CORTES	System and Technical Context
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Significance of the Study	1. DEJIERAY JADE R. DELAN 2. JOEY P. LUZONG	- Project Beneficiaries - Healthcare Impact
Scope & Delimitation	1. JERALD T. BARRUGA 2. RAQUELIN L. LEBIOS 3. KIELA MAE F. SALVACION	- Project Scope - Project Boundaries
Definition of Terms	All Members	Each member will contribute definitions relevant to their scope and role.

Chapter I – Introduction

1.1 Background of Study

In recent years, the integration of artificial intelligence into various systems has opened new possibilities and opportunities. When it comes to healthcare and patient care, the global standard nowadays is associated with Artificial Intelligence that provides decision-making support to Healthcare Professionals when it comes to personalized treatment. The old and traditional Clinical Management Systems (CMS) right now used in the Philippines, particularly in Provinces and rural healthcare facilities, often rely on static protocols. It only functions as record-keeping software. And it lacks the ability to provide actual assistance to our healthcare professionals to become more productive, and it may delay or limit access to individualized care, since the existing systems cannot synthesize patient history, laboratory results, and clinical guidelines into context-aware recommendations. Because their traditional Clinical Management System (CMS) does not have any decision-support intelligence features like AI. To address this gap by integrating semantic-driven AI for personalized medicine recommendations would help align local healthcare practices with emerging standards in clinical informatics and improve the timeliness and relevance of treatment decisions. The present study proposes CorText (The name of AI), a semantic-driven AI framework designed to deliver personalized medicine recommendations within a clinical management system.

1.2 Statement of the Problem

Main Problem:

There are many Clinical Management Systems (CMS) in the Philippines, especially in provincial and rural healthcare facilities, it only function primarily as record-keeping tools and lack integrated decision-support intelligence feature like AI. As a result, clinicians face inefficiencies in care delivery and missed opportunities for personalized treatment because existing systems cannot contextualize patient data and provide accurate medicine recommendations.

Specific Problems:

- Current Clinical Management System (CMS) lack semantic reasoning capabilities to synthesize heterogeneous patient data, such as Demographics, Medical history, Lab Results and more, into context-aware treatment recommendations.
- The absence of AI-driven recommendation features can increase clinician workload by requiring manual interpretation of data and cross-referencing with various guidelines.
- Usability and interface limitations in existing CMS can reduce the adoption and usage effectiveness among healthcare staff in resource-constrained facilities.
- Security and validation gaps (authentication, data integrity) in some CMS compromise trust in automated or semi-automated decision-support.
- There is limited empirical evaluation comparing personalized recommendations from AI-augmented systems with outputs from conventional CMS.

Research Problem

How can a semantic-driven AI framework be designed and implemented within a Clinical Management System to generate accurate, usable, and secure personalized medicine recommendations for clinicians in provincial and rural healthcare settings?

1.3 Objectives of the Study

General Objective:

To design and implement a semantic-driven AI framework within a Clinical Management System (CMS) that provides accurate, usable, and secure personalized medicine recommendations to support clinicians in provincial and rural healthcare facilities.

Specific Objectives:

- To develop an AI for Clinical Management System capable of providing decision-support to healthcare professionals.
 - To integrate a semantic-matching engine using BioBERT embedding model that aligns patient data with appropriate medicine profiles.
 - To train the AI with all the FDA-Approved Medicines, and to integrate a clinician-driven feedback-loop to make the AI more intelligent.
 - To ensure data privacy and compliance with Philippine Healthcare Standards.
 - To implement AI to empower healthcare professionals with AI-driven CMS that enhance clinical decision-making, streamline workflows, and deliver personalized patient care.
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1.4 Significance of Study

The findings of this study will offer practical benefits to multiple stakeholders. For patients, CorText aims to improve the relevance and safety of prescribed medicines by delivering context-aware recommendations. Clinicians will gain a time-saving decision-support tool that increases confidence and reduces manual workload. Healthcare facilities in provincial and rural areas will benefit from improved operational efficiency and standardized care processes. Policymakers and health system planners will receive empirical evidence to guide CMS modernization, while researchers and developers will gain a documented semantic model, implementation patterns, and evaluation methods that support further work in semantic AI for healthcare. Collectively, these contributions advance the alignment of local clinical practice with contemporary standards in clinical informatics and personalized medicine.

1.5 Scope & Delimitation

➤ Scope:

CorText is an AI decision-support upgrade for a Clinical Management System that generates personalized medicine recommendations. The prototype includes a frontend, backend, database, a clinical knowledge model (domain ontology), and a recommendation engine with a clinician-driven feedback loop. It uses only structured patient data (demographics, medical history, diagnoses, lab results, current medications, allergies, contraindications) and a curated subset of FDA-approved drugs mapped to the semantic model. Evaluation will measure accuracy (agreement with clinicians or guidelines), usability (standard tests and task metrics), and basic security (authentication, access control, data integrity). Testing is limited to a controlled pilot/staging environment, and clinicians must approve any treatment decisions.

➤ **Delimitation:**

The project will not attempt exhaustive coverage of all FDA-approved medicines. But rather, it will use a documented, curated subset relevant to the pilot. CorText is explicitly a decision-support system and will not autonomously prescribe or alter patient treatment without clinician approval. Processing unstructured clinical notes, medical imaging, and advanced natural language processing are out of scope for the core prototype; any experimental NLP features will be identified clearly and limited. Full regulatory certification, large-scale performance optimization, and production-grade penetration testing are beyond the study's timeframe and resources. These delimitations prioritize patient safety, human-in-the-loop governance, and feasibility for a pilot-scale implementation; methodological specifics (curated drug list, sample sizes, retraining cadence, and evaluation instruments and thresholds) are detailed in Chapter 3 and all datasets, clinician feedback, and model versions will be logged and documented for traceability.

1.6 Definition of Terms

- **CorText** - The prototype AI feature developed in this study. It is a semantic-driven recommendation engine integrated into a Clinical Management System to provide personalized medicine suggestions to clinicians.
 - It is a combination of two words, “*Core*” and “*Text*”. “*Core*” represents the heart or central intelligence of Clinical Management System, which has a main purpose of understanding and processing information. On the other hand, “*Text*” refers to the textual information of the patient and medicine, which will be embedded into vector number representation that the AI could read, analyze, and connect – to get the relevance of the data by doing calculation to create an accurate medicine recommendation. Altogether, “*Core + Text = CorText*” means “*The Core intelligence of Clinical Management System that understands and connects textual medical and clinical information.*”
- **Clinical Management System (CMS)** – Record-Keeping Software used by healthcare facilities to record patient encounters, manage clinical data, and support administrative and clinical workflows.
- **Domain Ontology** - A structured, machine-readable model (vocabulary and relationships) that represents clinical concepts (e.g., Patient, Medication, LaboratoryTest) and the rules that connect them for consistent interpretation by the system.
- **Semantic-driven AI** - An AI approach that combines machine learning with an explicit semantic model (ontology) so that recommendations use both learned patterns and formalized clinical knowledge.
- **Personalized Medicine Recommendations** - Context-aware suggestions about medicine as suggestions based on the individual patient's data (diagnoses, labs, medications, allergies, comorbidities).
- **Clinician-driven feedback-loop** – A machine learning mechanism where clinicians confirm, correct, or annotate AI suggestions. The feedback is recorded and will be used to train the AI to improve its knowledge.
- **Curated subset of FDA-approved medicines** - A documented, manageable selection of FDA-approved drugs chosen for the pilot based on relevance and data availability, mapped to the semantic model.
- **Structured patient data** - Patient information stored in defined fields and tables (e.g., demographics, coded diagnoses, lab values, medication records), not free-text notes or images.

- **Active diagnosis** - A diagnosis currently relevant to the patient's care episode, recorded with a date and status (e.g., confirmed or provisional).
- **Contraindication** - A clinical reason (condition, lab abnormality, allergy, or drug interaction) that makes a particular drug inappropriate or unsafe for a specific patient.
- **Accuracy (in recommendations)** - The degree to which CorText's suggestions agree with clinician judgments or established guideline-based standards, measured by metrics such as precision, recall, or agreement rate.
- **Usability** - The effectiveness, efficiency, and satisfaction with which clinicians can use the system, commonly assessed with instruments like the System Usability Scale (SUS) and task performance metrics.
- **Security (prototype scope)** - Measures implemented for the research prototype that protect patient data and system integrity, including authentication, role-based access control, logging, and basic data-integrity checks.
- **Decision-support feature** – A feature that aids clinicians by providing relevant information or recommendations to inform decisions while keeping the clinician responsible for final actions. It's the AI feature of the System.
- **Coding standards** – A standard medical code systems used so different systems understand the same data, for example ICD for diagnoses, LOINC for lab tests, and RxNorm for medicines.